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Topic1: Programming Assignments

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1. Reverse a String (LC 344)

Given a string, write a program to reverse the sequence of characters and return the reversed string. The solution should handle all characters correctly while maintaining the original character values.
Example: Input: "hello" → Output: "olleh"

2. Valid Palindrome (LC 125)

Given a string, determine whether it reads the same forward and backward after ignoring spaces, punctuation, and case differences. Return true if the string is a palindrome; otherwise, return false.
Example: Input: "madam" → Output: true

3. Count Vowels and Consonants (No LeetCode)

Given a string, count the total number of vowels and consonants present in it. Consider only alphabetic characters while performing the count.
Example: Input: "apple" → Output: Vowels = 2, Consonants = 3

4. To Lower Case (LC 709)

Given a string containing uppercase letters, convert all characters into lowercase and return the modified string.
Example: Input: "HELLO" → Output: "hello"

5. Convert Lowercase to Uppercase (No LeetCode)

Given a string containing lowercase letters, convert all characters into uppercase and return the updated string.
Example: Input: "world" → Output: "WORLD"

6. Find Length Without Built-in Function (No LeetCode)

Given a string, determine its length without using any built-in length function. Traverse the string manually and count the characters.
Example: Input: "coding" → Output: 6

7. Number of Segments in a String (LC 434)

Given a sentence, count the total number of words or segments separated by spaces. Consecutive spaces should not be counted as extra words.
Example: Input: "I love programming" → Output: 3

8. Remove Spaces from String (LC 1592 Inspired)

Given a string containing spaces, remove all spaces and return the updated string without changing the order of remaining characters.
Example: Input: "a b c d" → Output: "abcd"

9. Check if Two String Arrays are Equivalent (LC 1662 Inspired)

Given two strings, compare them character by character and determine whether they are equal. Return true if both strings are identical; otherwise, return false.
Example: Input: "abc", "abc" → Output: true

10. Concatenate Two Strings (No LeetCode)

Given two strings, combine them into a single string and return the result. The original order of characters must remain unchanged.
Example: Input: "Hello", "World" → Output: "HelloWorld"

11. First Unique Character / Character Frequency (LC 387 Inspired)

Given a string, count the frequency of each character and display how many times every character appears in the string.
Example: Input: "banana" → Output: b=1, a=3, n=2

12. Replace Spaces with Hyphens (LC 1592 Inspired)

Given a string containing spaces, replace every space character with a hyphen (-) and return the modified string.
Example: Input: "hello world" → Output: "hello-world"

13. Print Duplicate Characters (No LeetCode)

Given a string, identify and print all characters that occur more than once in the string. Each duplicate character should be printed only once.
Example: Input: "programming" → Output: "r g m"

14. Valid Anagram (LC 242)

Given two strings, determine whether one string is an anagram of the other. Two strings are anagrams if they contain the same characters with the same frequency.
Example: Input: "listen", "silent" → Output: true

15. First Non-Repeating Character (LC 387)

Given a string, find and return the first character that appears only once in the string. If no such character exists, return an appropriate value.
Example: Input: "swiss" → Output: "w"

16. Reverse Words in a String III (LC 557)

Given a sentence, reverse the characters of each word while preserving the original order of the words and spaces.
Example: Input: "Hello World" → Output: "olleH dlroW"

17. Remove Vowels from a String (LC 1119)

Given a string, remove all vowels (a, e, i, o, u) from it and return the resulting string.
Example: Input: "education" → Output: "dctn"

18. Toggle Case of Characters (No LeetCode)

Given a string, change all uppercase letters to lowercase and all lowercase letters to uppercase. Return the transformed string.
Example: Input: "HeLLo" → Output: "hEllO"

19. Find the Index of the First Occurrence in a String (LC 28)

Given two strings, determine whether the second string exists as a substring inside the first string. Return the index or indicate if it is not found.
Example: Input: "hello world", "world" → Output: Found

20. Find ASCII Value of Characters (No LeetCode)

Given a character or string, print the ASCII value corresponding to each character.
Example: Input: "A" → Output: 65

Topic2: Interview Question

Java Interview Questions

1. What is Java and why is it called a platform-independent programming language?
2. What are the main features of Java?
3. What is JVM in Java and what is its role?
4. What is JRE and why is it required to run Java programs?
5. What is JDK and how is it different from JRE?
6. What is Bytecode in Java and how is it generated?
7. Differentiate between JDK, JRE, and JVM.
8. What are the different data types available in Java?
9. What are operators in Java and what are their different types?
10. Explain the "Write Once, Run Anywhere" concept in Java.

TOPIC3: JAVA MCQ Questions

Java MCQ Questions

1. What is Java?

Which of the following best describes Java?

A. Database Management System
B. Operating System
C. Platform-independent programming language
D. Web browser

Answer: C. Platform-independent programming language

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2. Which feature makes Java platform independent?

A. Pointers
B. Bytecode and JVM
C. Header files
D. Assembly language

Answer: B. Bytecode and JVM

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3. What is the full form of JVM?

A. Java Variable Method
B. Java Visual Machine
C. Java Virtual Machine
D. Java Verified Machine

Answer: C. Java Virtual Machine

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4. What is the primary role of JVM?

A. Compile Java code
B. Store Java programs
C. Execute Java bytecode
D. Design GUI applications

Answer: C. Execute Java bytecode

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5. What does JRE stand for?

A. Java Runtime Environment
B. Java Ready Environment
C. Java Runtime Editor
D. Java Resource Environment

Answer: A. Java Runtime Environment

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6. Which component is required to develop Java applications?

A. JRE
B. JVM
C. JDK
D. JAR

Answer: C. JDK

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7. What is Bytecode in Java?

A. Machine-level code
B. Intermediate code generated by compiler

C. Source code written by programmer
D. Operating system code

Answer: B. Intermediate code generated by compiler

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8. Which of the following is NOT a primitive data type in Java?

A. int
B. float
C. String
D. char

Answer: C. String

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9. Which operator is used for comparison in Java?

A. =
B. ==
C. &&
D. ++

Answer: B. ==

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10. What does “Write Once, Run Anywhere” mean in Java?

A. Java code runs only on Windows
B. Java programs need recompilation for every OS
C. Java bytecode can run on any system with JVM
D. Java supports only one platform

TOPIC4: Aptitude Question

1. Student Marks Percentage

A student appears for an examination and scores marks out of the total marks available. Find the percentage of marks obtained by the student.
Example: A student scores 450 marks out of 600 marks. Find the percentage obtained.

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2. Percentage Increase in Price

The price of an item increases after some time. Calculate the percentage increase in the price.
Example: The price of a shirt increases from ₹800 to ₹920. Find the percentage increase.

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3. Percentage Discount

A shopkeeper offers a discount on the marked price of a product. Calculate the final price after applying the discount percentage.
Example: A product worth ₹2000 is available at a 15% discount. Find the final price.

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4. Percentage of Boys and Girls

In a class, some students are boys and the remaining are girls. Find the percentage of boys or girls in the class.
Example: In a class of 50 students, 30 are boys. Find the percentage of girls.

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5. Original Number from Percentage Increase

A number is increased by a certain percentage and the final value is given. Find the original number.
Example: A number is increased by 25% and becomes 500. Find the original number.

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6. Percentage Decrease in Population

The population of a town decreases over a period of time. Calculate the percentage decrease in population.
Example: The population decreases from 80,000 to 72,000. Find the percentage decrease.

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7. Salary Expense Percentage

A person spends a fixed percentage of salary on a particular expense. Calculate the amount spent.
Example: A person spends 40% of a ₹50,000 salary on rent. Find the rent amount.

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8. Passing Percentage in Examination

A student must score a minimum percentage to pass an exam. Determine whether the student passes or fails based on marks obtained.

Example: Passing marks are 35%. A student scores 280 marks out of 700. Check whether the student passes.

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9. Depreciation Percentage

The value of an object decreases every year by a fixed percentage. Find the previous or present value using the depreciation percentage.
Example: A machine depreciates by 10% annually and its current value is ₹90,000. Find its value last year.

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10. Election Vote Percentage

A candidate receives a certain percentage of total votes in an election. Calculate the number of votes received by the candidate.
Example: Candidate A receives 60% of 5000 votes. Find the number of votes received.